

Surface-Related Multiples as a Feedback System

1 Introduction

Surface-related multiples can be understood naturally using the language of *feedback systems*. The idea is simple: a primary reflection reaches the surface, is reflected back downward, interacts with the subsurface again, and therefore generates additional reflected arrivals. Repetition of this process creates higher-order multiples.

We use the following geophysical notation:

- w : source wavelet,
- P : primary reflection response, or primary reflection operator,
- R : reflection coefficient at the surface,
- M : multiple contribution to the data.

Throughout this note, $*$ denotes convolution.

2 Primaries Without Feedback

If there were no reflection at the free surface, the source wavelet w would propagate into the earth, interact with the subsurface reflectors, and generate the primary reflection response

$$d_P = P * w. \quad (1)$$

Thus, P represents the action of the subsurface on the incident wavelet.

3 The Surface as a Feedback Mechanism

When the primary wavefield reaches the surface, it is not necessarily removed from the system. Instead, it can be reflected back downward with reflection coefficient R .

The reflected wave then propagates through the earth again and interacts once more with the primary reflection operator P .

This produces the first-order multiple:

$$M_1 = P * R * P * w. \quad (2)$$

The physical sequence is therefore

$$w \longrightarrow P \longrightarrow R \longrightarrow P. \quad (3)$$

The first application of P creates the primary reflection. The factor R represents reflection at the surface, and the second application of P represents another interaction with the subsurface.

4 Higher-Order Multiples

After the first-order multiple reaches the surface, it can again be reflected downward. This produces a second-order multiple,

$$M_2 = P * R * P * R * P * w. \quad (4)$$

Similarly, the third-order multiple is

$$M_3 = P * R * P * R * P * R * P * w. \quad (5)$$

Each additional trip through the feedback loop introduces another factor $R * P$. The total recorded wavefield can therefore be written as

$$d = P * w + P * R * P * w + P * R * P * R * P * w + \dots. \quad (6)$$

The first term is the primary contribution, while the remaining terms are surface-related multiples.

Defining the total multiple contribution as M ,

$$M = P * R * P * w + P * R * P * R * P * w + \dots. \quad (7)$$

Hence,

$$d = P * w + M. \quad (8)$$

5 Operator Form

If convolution symbols are omitted for compactness, the data can be written as

$$d = Pw + PRPw + PRPRPw + \dots. \quad (9)$$

Factoring out the first primary operator gives

$$d = P [I + RP + (RP)^2 + (RP)^3 + \dots] w. \quad (10)$$

The expression inside brackets is a geometric operator series. Formally, when the series converges,

$$I + RP + (RP)^2 + \dots = (I - RP)^{-1}. \quad (11)$$

Therefore,

$$\boxed{d = P(I - RP)^{-1}w} \quad (12)$$

which has the characteristic form of a system with feedback.

6 Feedback Interpretation

The key physical interpretation is that the primary wavefield is *fed back* into the earth after reflecting at the surface.

A simplified feedback representation is shown in Figure 1.

Each traversal of the feedback path generates one additional order of multiple.

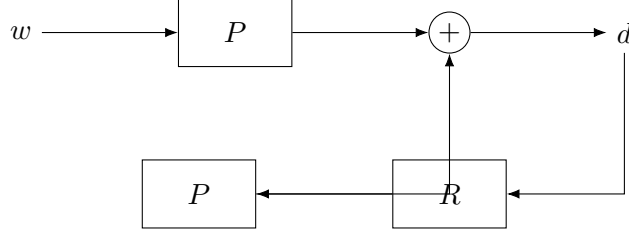


Figure 1: Conceptual representation of surface-related multiples as a feedback system. The wavefield reaching the surface is reflected by R and re-enters the earth, where it interacts again with the primary reflection operator P .

7 Role of the Surface Reflection Coefficient

For pressure waves at an ideal free surface, the reflection coefficient is often approximately

$$R \approx -1. \quad (13)$$

The negative sign represents the polarity reversal associated with reflection of pressure at the free surface.

Thus, the surface does more than merely return energy into the subsurface: it can also change the phase or polarity of that returned wavefield.

8 Main Interpretation

The feedback view provides a compact way to understand multiple generation:

A primary reflection reaches the surface. The surface reflects it back into the earth. The returned wavefield interacts again with the subsurface reflectors, creating a multiple. Repetition of this feedback process creates higher-order multiples.

In this interpretation,

$$\boxed{\text{primaries become the input to the feedback loop that generates multiples.}} \quad (14)$$

This viewpoint is useful because it connects geophysical multiple prediction with the general theory of systems with feedback. If the primary reflection operator P and the surface reflection operator R can be estimated, the multiple wavefield can in principle be predicted by repeatedly applying the feedback process.